

Lung Nodule Segmentation — DynUNet / 3D U-Net

1. Task

Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, cropped to a per-series lung bounding box.

- **Input** (1, 256, 256, 256) CT crop, intensity-normalised to [0, 1], resampled from the bbox (padding = 20 vox).
- **Output** (2, 256, 256, 256) softmax logits — class 0 = background, class 1 = nodule.

2. Model

[MONAI DynUNet](#) — nnU-Net-style **3D U-Net**, DynUNet (nnU-Net-style 3D U-Net).

Property	Value
Architecture family	DynUNet (nnU-Net-style 3D U-Net)
Spatial dims	3
Input channels	1
Output channels	2 (softmax)
Encoder levels	6
Kernel sizes	[3, 3, 3] at every level
Strides	[[1,1,1], [2,2,2]×5]
Normalisation	instance
Deep supervision	off
Trainable parameters	31,181,730

3. Data

Trained on the **unified** split (patient-grouped, dataset-stratified, **full corpus**: NLST + NSCLC + LIDC-IDRI).

Split sizes: **1 683 train / 297 val / 325 test (held out)**.

Class imbalance: nodule voxels are on the order of $\sim 10^{-5}$ of the total, which shapes how each metric should be read (§7).

4. Training

Setting	Value
Loss	Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1-0.3$, <code>ce_nodule_weight</code> = 100)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2
Epochs	400
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 × NVIDIA H100 94 GB
Wall-clock	≈ 3.5 days

Best checkpoint (evaluated below): **epoch 354 / 400**.

5. Evaluation protocol

Held-out test split (325 series). The training loop used the `val` split of `unified` for early stopping and best-Dice checkpoint selection; the numbers below come from the completely untouched `test` split (never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output; foreground = class 1. All metrics are computed at the voxel level and micro-averaged over the whole test split (5,452,595,200 voxels total).

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.7594
Accuracy	0.9993
Precision	0.6236
Recall	0.7569

Supplementary

Metric	Value
IoU (foreground / nodule class)	0.5196
IoU (background class)	0.9993
Dice / F1 (micro)	0.6838

Per-case Dice distribution (325 val cases)

Statistic	Value
Mean	0.5560
Std	0.2551
Min	0.0000
p05	0.0031
p25	0.3837
Median	0.6196
p75	0.7621
p95	0.8709
Max	0.9455

7. Notes on interpretation

Nodule segmentation is severely class-imbalanced ($\sim 10^{-5}$ of voxels are nodule), which distorts the standard metric set:

- **Accuracy** is trivially near 1.0 for any reasonable model — the model gets $\approx 99.9313\%$ of voxels right by predicting "background" almost everywhere. Uninformative on its own here.
- **mIoU** averages the foreground and background IoU. `IoU_background` is essentially 1.0, so mIoU is roughly $0.5 + 0.5 \cdot \text{IoU_nodule}$. The useful signal is in `IoU_nodule` (0.5196) and Dice / F1 (0.6838).
- **Precision / Recall** are the standard per-voxel figures — no imbalance caveat needed. Recall 0.7569 means the model correctly labels $\sim 76\%$ of nodule voxels; Precision 0.6236 means $\sim 62\%$ of predicted-nodule voxels are true nodule.

Per-case Dice mean (0.5560) is lower than the micro Dice (0.6838) because large nodules dominate the micro average; per-case Dice weights each patient equally regardless of nodule size. Clinically the per-case distribution is the more useful summary.

8. How to load & run inference

Pre-trained weights are hosted at [szabopeter/nodule-dynunet-3d](#). Download and load directly:

```
huggingface-cli download szabopeter/nodule-dynunet-3d --local-dir ./ckpt
```

```
import yaml, torch
from monai.networks.nets import DynUNet

cfg = yaml.safe_load(open("config.yaml"))["model"]
default_strides = [[1, 1, 1]] + [[2, 2, 2]] * 5
strides = cfg.get("strides", default_strides)
kernel = cfg.get("kernel_size", [[3, 3, 3]] * len(strides))
model = DynUNet(
    spatial_dims      = cfg["spatial_dims"],
    in_channels       = cfg["in_channels"],
    out_channels      = cfg["out_channels"],
    kernel_size       = kernel,
    strides           = strides,
    upsample_kernel_size = cfg.get("upsample_kernel_size", strides[1:]),
    norm_name         = cfg.get("norm_name", "instance"),
    deep_supervision  = cfg.get("deep_supervision", False),
)
state = torch.load("model.pth", map_location="cpu", weights_only=True)
model.load_state_dict(state)
model.eval()

with torch.no_grad():
    # Input: (B, 1, 256, 256, 256), lung-bbox-cropped CT resampled to 256³.
    x = torch.randn(1, 1, 256, 256, 256)
    logits = model(x)  # (B, 2, D, H, W)
    pred_class = logits.argmax(dim=1)  # (B, D, H, W) in {0, 1}
    nodule_mask = (pred_class == 1).to(torch.uint8)
```

This model expects **lung-bbox-cropped** input. A separate 2D ROI model is needed to produce that bbox — see the accompanying ROI checkpoints ([szabopeter/roi-segresnet-2d](#) or [szabopeter/roi-swinunetr-2d](#)), or use the demo's `inference.py` which chains them for you.

Instance-level metrics (per-nodule)

Connected-component analysis with 26-connectivity, pooled over the held-out test split, on the 256-cube resample grid (same grid as the voxel metrics above).

- **Instance recall** — GT nodules touched by at least one predicted voxel, over all GT nodules.
- **Instance precision** — predicted components touching at least one GT voxel, over all predicted components.

Metric	Value
GT nodules (components)	941
GT nodules hit	709
Instance recall	0.7535
Predicted components	1,644
Predicted components hitting GT	745
Instance precision	0.4532

No minimum-size filtering is applied — every predicted component counts, so single-voxel false positives lower instance precision. Produced by `scripts/instance_metrics.py --config configs/dynunet.yaml --ckpt <ckpt> --split test` (JSON: `results_instance/dynunet.json`).

9. Reproducibility

- Config `configs/dynunet.yaml`
- Checkpoint `checkpoints/dynunet/best_model.pth`
- Metrics JSON `checkpoints/dynunet/eval_metrics_nodule_test.json`
- Command `python eval_metrics_nodule.py --config configs/dynunet.yaml --ckpt checkpoints/dynunet/best_model.pth --split test`

Companion two-stage nodule models on HuggingFace:

- [szabopeter/nodule-segresnet-3d-small](#) — small SegResNet
- [szabopeter/nodule-segresnet-3d-wide](#) — wider SegResNet (paper's best two-stage)